Solution Specification Document for the Creation of Q-Blog on FMDQ Website

iQx Consult Limited

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SSD for the Creation of Q-Blog on FMDQ Website

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1 Acceptance and Confirmation

This Solution Specification Document ("**SSD**") serves as the official record of the requirements for this request. It is important to note that the contents of this document will be directly translated into test scripts for the system's quality assurance and user acceptance testing, to verify and validate the system's features and functionalities.

Any requirements or features not explicitly stated within this document will be regarded as out-of-scope and will be treated as a formal change request subject to the FMDQ application development framework.

By signing below, I do acknowledge and confirm that:

a. I have thoroughly reviewed and understood the requirements documented herein

b. The documented requirements fully represent my expectations and align with the objectives of the request/project

c. Any changes or additions to the documented requirements will be formally communicated and processed as change requests

2 Approvals

Document Name	SSD – Creation of Q-Blog on the FMDQ Website
Prepared By Oluwatoni Amusan	---------- **Business Interface Group** **May 2026**
Reviewed By Abbie Famuyide	---------- **Group Head, Business Interface** **May 2026**
Concurred By Oladayo Olujobi	---------- **Marketing Group** **May 2026**
Concurred By Nkiru Umeh	---------- **Group Head, Marketing** **May 2026**
Approved By Emmanuel Alao	---------- **Supervising Head, iQx Consult** **May 2026**

3 INTRODUCTION

The Marketing Group ("**MAG**") of FMDQ Group PLC ("**FMDQ**" or the "**Group**") has identified a need to strengthen its digital presence and deepen community engagement with its diverse stakeholder community. In response to this need, MAG has initiated the development of Q-Blog; a digital thought leadership and market insights blog to be hosted on the FMDQ Group website. This document specifies the technical and functional solution required to design, build, integrate, and deploy the Q-Blog in alignment with the requirements outlined in the approved Business Requirements Document ("**BRD**").

Q-Blog is designed to address a clear gap in how FMDQ currently communicates with the broader financial markets ecosystem. Market participants, regulators, academics, media professionals, students, and internal stakeholders have limited access to a single, structured, and authoritative platform through which FMDQ shares its knowledge and perspectives. This solution will provide these user groups with a centralised, accessible, and content-rich blog platform that delivers well-researched articles, regulatory commentary, investor education materials, and market insights, all organised under defined content categories and presented in a format that prioritises clarity and readability.

Q-Blog will enable FMDQ to consistently publish high-value content, improve financial literacy across its target audience, and drive organic traffic to the broader FMDQ digital ecosystem. The solution will empower the Group to manage content independently through a structured Content Management System ("**CMS**"), supported by a defined editorial workflow that ensures quality and institutional alignment before any article is made publicly available.

3.1 Purpose

This purpose of this document is to define the technical and functional specifications required for the design, development, and deployment of Q-Blog on the FMDQ Group website. It serves as the guiding reference for the implementation team to deliver a solution that meets the approved business requirements. Q-Blog is designed to provide FMDQ Group with a digital platform for sharing thought leadership, market insights, and educational content with its stakeholder community. The solution enables MAG to publish, manage, and distribute authoritative financial market content.

3.2 Scope

The scope of this Project includes the build-out of a public-facing blog platform with structured content categorisation, search and discovery functionality, social sharing capabilities, and a content archive. The solution will also include an internal CMS with role-based access control, and an approval workflow to support MAG in independently creating and publishing content, alongside the integration of Google Analytics 4 ("**GA4**") for traffic and engagement monitoring, as well as a native analytics dashboard to support performance reporting.

The following are explicitly outside the scope of this project:

a. a blog-specific newsletter digest
b. podcast integration
c. any features beyond what is defined in the approved BRD dated April 2026

These may be considered for future phases of the platform.

3.3 References

This document references the approved BRD for the Creation of Q-Blog on the FMDQ Website by MAG.

3.4 Assumptions and Constraints

Assumptions

- FMDQ Group's existing website infrastructure can accommodate the Q-Blog as an integrated section without requiring a full website rebuild or migration
- MAG will be responsible for all content review and publication approvals. iQx will not be responsible for content sourcing or quality control
- Author profiles to be created on the platform will be limited to VP-level staff and above, in line with FMDQ's communication policy. Any exceptions will require prior GCOO approval before the profile is created on the system
- FMDQ Group already holds an active Google Analytics 4 ("**GA4**") account, and the necessary access credentials will be made available to the development team for integration purposes
- All images, multimedia assets, and content to be uploaded at launch will be provided by MCD in the required formats and dimensions ahead of the go-live date

Constraints

- The platform must comply with all applicable data privacy standards with respect to the collection and storage of newsletter subscriber information
- All articles must pass through the defined editorial approval workflow before publication. The system must not permit any article to bypass the approval process and go live directly
- Network availability is a prerequisite for system accessibility

3.5 Document Overview

This SSD provides a detailed description of the functional and technical requirements, system architecture, and operational standards for the Q-Blog. It is structured in two parts: **Part 1** covers Functional Requirement Specifications, detailing user requirements, system modules, and process workflows; while **Part 2** covers Technical Requirements, addressing data architecture, interface standards, security and privacy controls, and operational guidelines. The SSD serves as the definitive implementation blueprint for the implementation team.

4 METHODOLOGY

The major goal is to provide a system that delivers a reliable, structured, and content-rich blog for FMDQ Group, empowering the MAG to publish and manage thought leadership content efficiently while providing external stakeholders with seamless access to market insights and educational materials. The system will be primarily administered and managed by MAG but will have different administrator categories to support governance and operational control. The various administrator categories are as follows:

a. MAG Super Admin

b. MAG Admin

The system will have modules to cater for content creation and publishing, content categorisation and discovery, search and archive functionality, and analytics and performance reporting. Through a structured approval workflow, role-based access controls, and integration with Google Analytics 4, the system will streamline content governance and ensure that only approved, high-quality articles are made publicly available. Ultimately, the platform aims to strengthen FMDQ's digital presence, improve financial literacy among its stakeholder community, and position the organisation as the leading voice in Nigeria's financial markets.

5 FUNCTIONAL REQUIREMENTS

5.1 Context - Generic Context Table

The proposed system to consists of the following major actors and their activities:

S/N	ACTORS	ACTIVITIES
1.	MAG Admin	▪ Create and manage author profiles on the platform ▪ Modify author profiles (deactivate/reactivate) ▪ Review and approve articles submitted by authors before publication ▪ Publish, unpublish, edit, and archive articles on the blog ▪ Manage and define content categories, tags, and featured article selections ▪ Monitor blog performance via the internal analytics dashboard
2	MAG Super Admin	▪ Authorise (approval or rejection to deactivate/reactivate) user accounts and roles
3	Article Authors	▪ Create and draft blog articles using the CMS rich-text editor ▪ Upload images and embed multimedia assets within articles ▪ Save articles as drafts ▪ Preview articles before submission ▪ Submit articles for review and approval
4	Public users (External)	▪ Browse and read published blog articles on the Q-Blog landing page ▪ Search and filter articles by category, keyword, or author ▪ Share articles across social media platforms ▪ Download, print, or copy article links ▪ Subscribe to the FMDQ newsletter via the subscription form

5.2 User Requirements

The following are high level requirements of the proposed system, considering all major classes/categories of user:

- The system shall provide a rich-text editor for authors to create, format, and manage blog articles
- The system shall allow authors to save articles as drafts, preview them before submission, and submit them for approval
- The system shall allow MCD Admin to manage and update content categories and tags from within the CMS
- The system shall allow public users to filter articles by category and sort by most recent or most read
- The system shall provide full article pages displaying all metadata including title, author, publication date, category label, estimated reading time, and featured image
- The system shall provide social sharing options for LinkedIn, X (Twitter), Facebook, and Email on every article page
- The system shall integrate with Google Analytics 4 (GA4) for traffic and engagement monitoring

- The system shall provide an internal CMS-native analytics dashboard for on-platform performance tracking
- The system shall support the export of analytics data in standard formats such as CSV and PDF
- The system shall enforce role-based access control with distinct permissions for the MAG Super Admin, MAG Admin, and Author roles

5.3 System Utilisation

The following represents the estimated system utilisation for the Q-Blog based on the identified user categories. These figures are assumptions made at the time of documentation:

S/N	User Category	Estimated Total Users	Estimated Concurrent Users
1	MAG Super Admin	1	1
2	MAG Admin	5	2
3	Article Authors	10	3
4	Public Users	5,000	250
Total		**5,016**	**256**

The platform must be designed to comfortably support the estimated concurrent user load stated above, with sufficient headroom to accommodate growth in public readership as the Q-Blog gains traction and visibility. Traffic spikes beyond the estimated concurrent user figures, particularly during periods of major market events, regulatory announcements, or content campaigns must also be accounted for in the architecture of the solution.

Commented [KA1]: Regarding infrastructure scaling driven by increased traffic, do we have a pre-approved process that takes financial considerations into account as well, and not just the technical requirements? I would like to understand whether budget approval and cost impact are already factored into the scaling process.

Commented [OA2R1]: Kindly engage ITD for more information on this.

5.4 Functional Requirements

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
1.	**User Profiling**	Automatic Author Profiling	As a system, I want to automatically profile all FMDQ staff at VP grade level and above as Authors on the CMS, So that eligible staff are readily available on the platform without requiring manual one-by-one profiling.	**Automatic Profile Creation** ▪ Given that a staff member at VP grade level or above exists in FMDQ ▪ When the system is being built ▪ Then the staff member shall be automatically profiled as an Author on the CMS with the appropriate role and access permissions assigned ▪ And the necessary details (a list of users to be profiled, their names and email addresses) shall be provided by MAG to iQx **Exceptional Profiling** ▪ Given that a staff member below VP grade level is to be profiled as an Author by exception ▪ When the MAG Super Admin is informed ▪ An interface shall be made available for the MAG Admin to initiate creation manually and assign a role to the user ▪ And it shall be routed to the MAG Super Admin for approval ▪ Then the MAG Super Admin shall be able to approve or reject the profile manually on the CMS ▪ And if the MAG Super Admin approves ▪ Then the profile shall immediately become created, with the role assigned, on the system ▪ And if the MAG Super Admin rejects ▪ They shall input a reason for rejection ▪ And the action shall be rejected **Staff Exit or Grade Change** ▪ Given that a profiled Author exits the organisation ▪ When the MAG Admin deactivates the profile

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ It shall be routed to the MAG Super Admin for approval ▪ Then the Author's access to the CMS shall be immediately revoked and their profile shall be marked as inactive without affecting any previously published articles attributed to them
		CMS Menu Access (Author)	As a profiled Author, I want to have access to the relevant menu items on the CMS, So that I can manage my articles and personal profile from a single interface.	**Author Menu Availability** ▪ Given that I am logged in as a profiled Author ▪ When I view the CMS navigation ▪ Then the following menu items shall be visible and accessible to me: a. My Articles b. Profile
		CMS Menu Access (MAG Admin)	As an MAG Admin, I want to have access to all relevant menu items on the CMS, So that I can manage my own articles, oversee all platform content, and maintain my personal profile.	**MAG Admin Menu Availability** ▪ Given that I am logged in as an MAG Admin ▪ When I view the CMS navigation ▪ Then the following menu items shall be visible and accessible to me: a. My Articles b. All Articles c. Profile **MAG Admin Full Access** ▪ Given that I am logged in as an MAG Admin ▪ When I navigate through the CMS ▪ Then I shall have unrestricted access to all CMS menu items and features available on the platform without any access restrictions
		Profile Management	As a profiled Author or MAG Admin,	**Profile Menu Access** ▪ Given that I am logged in to the CMS ▪ When I click on the "Profile" menu item

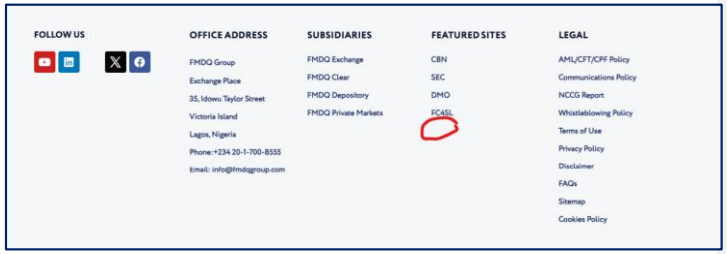
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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
			I want to be able to view and manage my profile on the CMS, So that my personal information is accurate and my account remains secure	▪ Then my profile page shall load displaying my name and available account management options **Profile Name Display** ▪ Given that I am on my profile page ▪ When the page loads ▪ Then my full FMDQ name as registered on the CMS shall be clearly displayed on my profile page **Password Reset** ▪ Given that I am on my profile page ▪ When I select the option to reset or update my password ▪ Then the system shall prompt me to enter my current password followed by new password and confirmation of the new password before the change is saved **Password Policy Compliance** ▪ Given that I am updating my password on my profile page ▪ When I enter a new password that does not comply with FMDQ's password policy ▪ Then the system shall display a clear validation error describing the password requirements and the password shall not be updated until a compliant password is entered **Successful Password Update** ▪ Given that I have entered a current password, a compliant new password, and a matching confirmation password ▪ When I submit the password update ▪ Then the system shall save the new password, display a confirmation message, and require me to use the new password on my next login
2.		Menu Navigation	As a public user,	**Menu Placement**

Commented [KA3]: Is MFA enabled for Admin login or only basic Authentication is enabled?

Commented [OA4R3]: This is not included in the SSD, given that the implementation team follows standard practice and procedure.

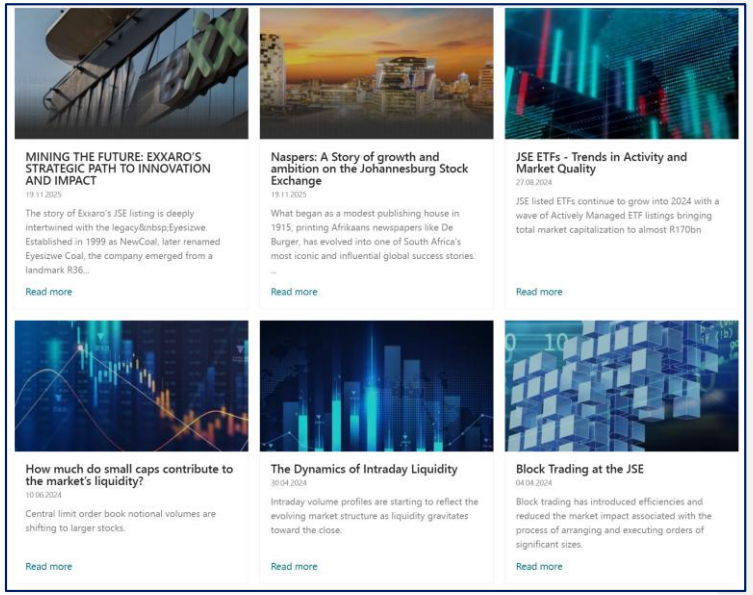
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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
	Website Navigation		I want to access the Q-Blog from the FMDQ website navigation menu, So that I can easily find and visit the blog from any page on the website.	▪ Given that I am on the FMDQ website ▪ When I look at the main navigation menu ▪ Then I shall see a hyperlink titled "FMDQ Q-Blog" placed beside the "FMDQ Next" navigation item  **Navigation Click** ▪ Given that I can see the "FMDQ Q-Blog" hyperlink in the main navigation menu ▪ When I click on it ▪ Then I shall be directed to the Q-Blog landing page without errors
		Footer Navigation	As a public user, I want to access the Q-Blog from the footer of the FMDQ website, So that I can navigate to the blog from any point on the website regardless of where I am on the page.	**Footer Placement** ▪ Given that I am on any page of the FMDQ website ▪ When I scroll to the footer ▪ Then I shall see a hyperlink titled "FMDQ Q-Blog" listed as a footer navigation item on every instance of the footer across the website ▪ And this shall be added to the list of 'Featured Sites' **Kindly refer to the screenshot below:** 

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Footer Click ▪ Given that I can see the "FMDQ Q-Blog" hyperlink in the footer ▪ When I click on it ▪ Then I shall be directed to the Q-Blog landing page
		Breadcrumb Navigation	As a public user, I want to see a breadcrumb trail as I navigate through the Q-Blog, So that I can easily track where I am within the website and navigate back to a previous page if needed.	**Landing Page Breadcrumb** ▪ Given that I have navigated to the Q-Blog landing page ▪ When the page loads ▪ Then the breadcrumb trail shall display the correct page hierarchy reflecting my current location on the website (e.g., Home > Q-Blog) **Article Page Breadcrumb** ▪ Given that I have clicked on an article and navigated to the article page ▪ When the page loads ▪ Then the breadcrumb trail shall update to reflect the correct page hierarchy (e.g., Home / Q-Blog / Article Title) **Kindly refer to the screenshot below:**  **Breadcrumb Navigation** ▪ Given that I am on an article page, and I can see the breadcrumb trail ▪ When I click on any item in the breadcrumb trail ▪ Then I shall be directed to the corresponding page without any errors ▪ And this shall enable me to track where I am on the website

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
3.	**Landing Page**	Article Display	As a public user, I want to see the six (**6**) most recently published articles on the Q-Blog landing page, So that I can quickly access the latest content without having to search for it.	**Article Display on Landing Page** ▪ Given that I have navigated to the Q-Blog landing page ▪ When the page loads ▪ Then the six (**6**) most recently published articles shall be displayed in chronological order with the most recent article appearing first **Kindly refer to the screenshot below:**  **Article Card Display** ▪ Given that the six (**6**) most recent articles are displayed on the landing page ▪ When I view each article card ▪ Then each card shall display the following details: a. featured image

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				b. article category c. article title **d.** "Read More" hyperlink **Article Card Click** ▪ Given that I can see the article cards on the landing page ▪ When I click anywhere on an article card or the "Read More" hyperlink ▪ Then I shall be directed to the full article page which shows the full details of the article
		View All Articles	As a public user, I want to be able to view all published articles on the Q-Blog, So that I can browse the full content library beyond the six articles displayed on the landing page.	**View All Button & Hyperlink Display** ▪ Given that I am on the Q-Blog landing page ▪ When the page loads ▪ Then a "View All Articles" button shall be visible at the bottom of the six displayed article cards ▪ And a "View All Articles" hyperlink shall be at the top of the six displayed article cards **View All Button/Hyperlink Click** ▪ Given that I can see the "View All Articles" button or hyperlink ▪ When I click on it ▪ Then I shall be directed to a page displaying all published articles on the blog **Default Sort Order** ▪ Given that I have clicked "View All Articles" and the full article list has loaded ▪ When I view the list ▪ Then articles shall be displayed in descending order by publication date by default with the most recently published article appearing first

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
		Sort	As a public user, I want to be able to sort the list of all articles, So that I can organise the content in a way that is most convenient for me.	**Sort Options Availability** ▪ Given that I am on the "View All Articles" page ▪ When I interact with the sort option ▪ Then I shall see the following sort options: a. publication date ascending b. publication date descending c. article name ascending d. article name descending e. category ascending f. category descending **Sort by Publication Date** ▪ Given that I am on the "View All Articles" page ▪ When I select to sort by publication date in ascending or descending order ▪ Then the article list shall re-order accordingly **Sort by Article Name** ▪ Given that I am on the "View All Articles" page ▪ When I select to sort by article name in ascending or descending order ▪ Then the article list shall re-order alphabetically in the selected order **Sort by Category** ▪ Given that I am on the "View All Articles" page ▪ When I select to sort by category in ascending or descending order ▪ Then the article list shall re-order by category name in the selected order
		Filter	As a public user,	**Filter Options Availability** ▪ Given that I am on the "View All Articles" page ▪ When I interact with the filter option

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
			I want to be able to filter the list of articles, So that I can narrow down the content to what is most relevant to my interests.	▪ Then I shall see filter options for content category, category label, author, and date range **Kindly refer to the screenshot below:**  **Filter by Content Category** ▪ Given that I am on the "View All Articles" page ▪ When I select a content category from the filter options ▪ Then only articles belonging to that content category shall be displayed and all other articles shall be hidden **Filter by Category Label** ▪ Given that I am on the "View All Articles" page ▪ When I select at least one category label from the filter options ▪ Then only articles tagged with that category label shall be displayed with no page errors ▪ And all other articles shall be hidden **Filter by Author** ▪ Given that I am on the "View All Articles" page ▪ When I select one or more authors from the filter options ▪ Then only articles written by that author shall be displayed **Filter by Author** ▪ Given that I am on the "View All Articles" page ▪ When I select the date range from the filter options ▪ Then only articles within that range of publication date shall be displayed

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Combined Filter ▪ Given that I am on the "View All Articles" page ▪ When I select one or more filter options ▪ Then the system shall be smart enough to display only applicable articles to the filter options **Clear Filter** ▪ Given that I have applied one or more filters ▪ When I clear or reset the filter selections ▪ Then the full list of all published articles shall be restored and displayed in the default sort order (descending order by publication date, i.e., most recent published articles comes first)
		Search	As a public user, I want to search for articles using keywords, So that I can quickly find content that is relevant to what I am looking for.	**Search Bar Visibility** ▪ Given that I am on the Q-Blog landing page or the "View All Articles" page ▪ When the page loads ▪ Then a clearly visible search bar shall be displayed on the page **Keyword Search** ▪ Given that I can see the search bar ▪ When I type a keyword and submit my search ▪ Then the system shall return all articles whose title, content, author, or tags match the entered keyword **No Results Found** ▪ Given that I have submitted a keyword search ▪ When no articles match the entered keyword ▪ Then the system shall display a clear message informing me that no results were found for my search query

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Search Context Retention ▪ Given that I have performed a search ▪ And clicked on an article from the search results ▪ When I navigate back to the search results page ▪ Then my previous search query and results shall be retained without requiring me to re-enter the keyword ▪ Except I empty or clear the search bar
4.	**Detailed Article Page**	Article Access	As a public user, I want to be directed to the full article page when I click on an article card, So that I can read the complete content of the article.	**Access from Landing Page** ▪ Given that I am on the Q-Blog landing page ▪ When I click on an article card or the "Read More" hyperlink ▪ Then I shall be directed to the full article page for that article **Access from View All Articles** ▪ Given that I am on the "View All Articles" page ▪ When I click on an article card ▪ Then I shall be directed to the full article page for that article **Access from Search Results** ▪ Given that I have performed a search and results are displayed ▪ When I click on an article card from the search results ▪ Then I shall be directed to the full article page for that article
		Article Metadata	As a public user, I want to see the article's metadata at the top of the article page, So that I can quickly identify key details about the article before I begin reading.	**Metadata Display** ▪ Given that I have navigated to a full article page ▪ When the page loads ▪ Then the following metadata shall be displayed at the top of the page: a. article title b. author name c. publication date

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				d. category label e. estimated reading time **Reading Time Calculation - Text Only** ▪ Given that an article has been created and submitted for publication ▪ When the system wants to calculate the estimated reading time ▪ Then the system shall divide the total word count of the article by 265 words per minute ("**WPM**") and translate the result into minutes and seconds (e.g., a 600-word article shall display an estimated reading time of approximately 2 minutes and 15 seconds) **Reading Time Calculation - (Contains Images)** ▪ Given that an article contains embedded images ▪ When the system calculates the estimated reading time ▪ Then the system shall adjust the reading time by: a. adding 12 seconds for the first image b. adding 11 seconds for the second image c. and subtracting 1 additional second for each subsequent image up to the tenth image, with no further adjustment applied beyond the tenth image **Metadata Accuracy** ▪ Given that an article has been published with all required metadata fields completed in the CMS ▪ When I view the article page ▪ Then all metadata fields shall be accurately populated and visible at the top of the page

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
		Article Formatting	As a public user, I want the article body to be clearly formatted and structured, So that I can read and navigate the content comfortably.	**Text Formatting** ▪ Given that I am on a full article page ▪ When the page loads ▪ Then the article body shall correctly render all text formatting including headings, subheadings, bullet points, and hyperlinks as originally structured in the CMS **Embedded Media** ▪ Given that an article contains embedded images, charts, infographics, or other visual assets ▪ When the page loads ▪ Then all visual assets shall be rendered correctly within the article body, display responsively across all device types, and load without disrupting the readability of the article content **Hyperlinks** ▪ Given that an article contains hyperlinks within its body ▪ When I click on a hyperlink ▪ Then I shall be directed to the correct destination Uniform Resource Locator ("**URL**") and external links shall open in a new browser tab without navigating me away from the article page
		Share	As a public user, I want to be able to share an article with my network, So that I can distribute content I find valuable across my preferred social media platforms or via email.	**Share Button Visibility** ▪ Given that I am on a full article page ▪ When the page loads ▪ Then a share button shall be clearly visible at the top of the article page **Share Popup Display** ▪ Given that I can see the share button at the top of the article ▪ When I click on the share button ▪ Then a popup interface shall appear displaying the following sharing options:

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				a. Copy Link b. LinkedIn c. X (Twitter) d. Facebook e. Instagram f. Email **Kindly refer to the screenshot below:**  **Social Platform Share** ▪ Given that the share popup is displayed ▪ When I click on any of the social platform options (LinkedIn, X, or Facebook) ▪ Then the corresponding social media sharing interface shall open with the article link pre-populated and ready to share **Email Share** ▪ Given that the share popup is displayed ▪ When I click on the Email sharing option ▪ Then my default email client shall open with the article link pre-populated in the email body ready to be sent

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S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Close Popup ▪ Given that the share popup is displayed ▪ When I click outside the popup or on a close button ▪ Then the popup shall be dismissed
		Download	As a public user, I want to download an article as a PDF, So that I can save and read the content offline at my convenience.	**Download Button Visibility** ▪ Given that I am on a full article page ▪ When the page loads ▪ Then a download button shall be clearly visible near the share button at the top of the article page **PDF Download** ▪ Given that I can see the download button ▪ When I click on it ▪ Then the system shall act just as if the user entered 'Ctrl + P' on their device, then generate and download the article as a PDF file, preserving the article's formatting, metadata, images, and content as displayed on the page **PDF Content Accuracy** ▪ Given that the PDF has been downloaded ▪ When I open the downloaded file ▪ Then the PDF shall accurately reflect the full article content including the title, author, publication date, category label, estimated reading time, body text, and all embedded visual assets
		Related Articles	As a public user, I want to see related articles at the end of the article I am reading,	**Related Articles Display** ▪ Given that I have navigated to a full article page ▪ When I scroll to the end of the article ▪ Then a section displaying related article cards shall be visible at the bottom of the page

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
			So that I can continue exploring content that is relevant to my interests.	**Related by Category/Author** ▪ Given that related articles are displayed at the bottom of the article page ▪ When I view the related articles section ▪ Then the displayed articles shall include articles that belong to the same content category or articles written by the same author as the article I am currently reading **Related Article Card Details** ▪ Given that related articles are displayed at the bottom of the article page ▪ When I view each related article card ▪ Then each card display shall be the same details as article cards on the landing page including featured image, article category, article title, a "Read More" hyperlink **Related Article Navigation** ▪ Given that I can see the related article cards at the bottom of the page ▪ When I click on a related article card or its "Read More" hyperlink ▪ Then I shall be directed to the full article page for that related article
5.	Content Categorisation	Category Assignment	As an Author, I want to assign at least one content category to my article, So that readers can easily identify the type of content they are reading.	**Category Selection** ▪ Given that I am creating or editing an article in the CMS ▪ When I attempt to submit the article for review ▪ Then the system shall require that at least one of the following predefined content categories has been selected before submission is permitted: Market & Economy, Innovation & Trends, Thought Leadership, Investor Education, Sustainability, Risk & Compliance, Product Updates, and Policy & Regulations.

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Category Not Selected ▪ Given that I am creating or editing an article in the CMS ▪ When I attempt to submit the article for review without selecting a content category ▪ Then the system shall display a clear validation error message instructing me to select at least one category before proceeding and the submission shall not be completed. **Multiple Category Selection** ▪ Given that I am creating or editing an article in the CMS ▪ When I select more than one content category for an article ▪ Then the system shall accept and save all selected categories and display all assigned category labels on the article card and article page.
		Category Label Display	As a public user, I want to see the category label on every article card and article page, So that I can quickly identify the type of content an article covers before deciding to read it.	**Category Label on Article Card** ▪ Given that articles are displayed on the landing page, the "View All Articles" page, or search results ▪ When I view an article card ▪ Then the assigned category label shall be clearly visible on the article card. **Category Label on Article Page** ▪ Given that I have navigated to a full article page ▪ When the page loads ▪ Then the assigned category label shall be displayed as part of the article metadata at the top of the page. **Multiple Category Labels** ▪ Given that an article has been assigned more than one content category ▪ When I view the article card or article page

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then all assigned category labels shall be displayed visibly without any label being omitted
		Category Management	As the MAG Admin, I want to be able to create, edit, and deactivate content categories in the CMS, So that the category list remains relevant and aligned with FMDQ's content strategy over time.	**Create Category** ▪ Given that I am logged in as an MAG Admin and I am on the category management section of the CMS ▪ I shall have a view of all content categories with the option to create new ones ▪ When I create a new content category and save it ▪ Then the new category shall appear in the category selection list available to authors when creating or editing an article **Edit Category** ▪ Given that I am logged in as an MAG Admin and I am on the category management section of the CMS ▪ When I edit the name or description of an existing content category and save the changes ▪ Then the updated category name shall reflect immediately across all article cards, article pages, and filter options where that category is displayed **Deactivate Category** ▪ Given that I am logged in as an MAG Admin and I am on the category management section of the CMS ▪ When I deactivate an existing content category ▪ The system shall prompt me to input my reason for deactivating ▪ Then the category shall no longer appear as an option in the category selection list for new articles, but all previously published articles assigned to that category shall retain their category label without disruption

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				Predefined Categories ▪ Given that the Q-Blog platform has been deployed ▪ When I log in as an MAG Admin and navigate to the category management section ▪ Then all eight predefined content categories shall already be available in the system: a. Market & Economy b. Innovation & Trends c. Thought Leadership d. Investor Education e. Sustainability f. Risk & Compliance g. Product Updates h. Policy & Regulations
6.	**Content Management System ("CMS")**	CMS Access	As a profiled Author or MAG Admin, I want to be able to log in to the CMS, So that I can access the content management features available to my role.	**Authorised Access: Author or MAG Admin** ▪ Given that I am a profiled Author or MAG Admin ▪ When I log in to the CMS with my valid credentials ▪ Then I shall be granted access to the CMS and shown the interface and menu options applicable to my role **Unauthorised Access** ▪ Given that I am a user who has not been profiled as an Author or MAG Admin ▪ When I attempt to log in to the CMS ▪ Then the system shall deny me access and display a clear error message informing me that I am not authorised to access the platform

SSD for the Creation of Q-Blog on FMDQ Website

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
		Rich-Text Editor	As a profiled Author or MAG Admin, I want to use a rich-text editor to write and format my articles so that I can produce well-structured and visually appealing content without requiring technical knowledge.	**Editor Availability** ▪ Given that I am logged in to the CMS as a profiled Author or MAG Admin ▪ When I navigate to the article creation page ▪ Then a rich-text editor shall be available and ready for use **Text Formatting** ▪ Given that I am writing an article in the rich-text editor ▪ When I apply formatting options ▪ Then the editor shall correctly support and render headings, subheadings, bullet points, numbered lists, bold, italics, underline, and hyperlinks within the article body **Image Upload** ▪ Given that I am writing an article in the rich-text editor ▪ When I upload an image ▪ Then the image shall be embedded within the article body at the desired position and display correctly in both the editor and the article preview ▪ And alt-text fields shall be available for images so as to improve accessibility and SEO **Multimedia Embedding** ▪ Given that I am writing an article in the rich-text editor ▪ When I embed multimedia content such as charts, infographics, or videos ▪ Then the multimedia content shall be embedded correctly within the article body and render as expected in both the editor and the article preview ▪ The acceptable file types are: a. **Images**: JPEG/JPG, PNG, WebP, GIF (static only) and SVG – these should have alt-text fields b. **Documents**: PDF

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				c. **Infographics/Charts**: JPEG/JPG, PNG and SVG d. **Video (if embedded)**: MP4 or Via external embed links (e.g., YouTube) ▪ And the file size limits are: a. **Images**: maximum 5MB per file b. **Videos**: maximum 20MB per file (if direct upload is supported)
		Preview	As a profiled Author or MAG Admin, I want to preview my article before saving or publishing it, So that I can see exactly what the finished article will look like to a public reader before it goes live.	**Preview Button Visibility** ▪ Given that I am writing or editing an article in the CMS ▪ When I view the article creation or editing page ▪ Then a "Preview" button shall be clearly visible alongside the other action buttons **Preview Accuracy** ▪ Given that I am writing or editing an article in the CMS ▪ When I click the "Preview" button ▪ Then the system shall display an accurate representation of the supposed final published layout of the article, including all formatting, metadata, images, and embedded media as they would appear to a public reader **Preview Does Not Publish** ▪ Given that I have clicked the "Preview" button and I am viewing the article preview ▪ When I close the preview ▪ Then the article shall not be published as a result of clicking the preview button ▪ And my article shall be saved as a draft content
		Save as Draft	As a profiled Author or MAG Admin, I want to save my article as a draft,	**Save as Draft Button Visibility** ▪ Given that I am writing or editing an article in the CMS ▪ When I view the article creation or editing page ▪ Then a "Save as Draft" button shall be clearly visible alongside the other action buttons

SSD for the Creation of Q-Blog on FMDQ Website

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
			So that I can return to it later to continue writing before submitting it for review or publishing it.	▪ And when I click the "Save as Draft" button ▪ Then the article shall be saved as a draft ▪ And a confirmation message shall be displayed ▪ Then the article shall appear in my "Drafts" tab under "My Articles" **Draft Re-opened** ▪ Given that I have previously saved an article as a draft ▪ When I navigate to my "Drafts" tab and click on the saved draft ▪ Then the article shall open in the rich-text editor with all previously saved content intact and ready for me to continue editing **Draft Not Published** ▪ Given that I have saved an article as a draft ▪ When I check the public-facing Q-Blog ▪ Then the draft article shall not be visible or accessible to public users until I publish it
		Publish	As an MAG Admin, I want to publish an article immediately upon clicking the "Publish" button, So that approved contents can be made available to the public without requiring additional approval steps.	**Publish Button Visibility** ▪ Given that I am logged in as an MAG Admin ▪ And I am writing or editing an article ▪ When I view the article creation or editing page ▪ Then a "Publish" button shall be clearly visible alongside the other action buttons **Immediate Publication: MAG Admin** ▪ Given that I am logged in as an MAG Admin and I have completed writing an article ▪ When I click the "Publish" button ▪ Then the article shall be immediately published and made publicly visible on the Q-Blog without requiring any further approval

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ And the article shall appear in the "Published" tab under "My Articles" **Article Routed for Approval: Author** ▪ Given that I am logged in as a profiled Author and I have completed writing an article ▪ When I click the "Publish" button ▪ Then the article shall not be immediately published but shall instead be routed to the MAG Admin for review and approval ▪ And the article shall appear in my "Pending Approval" tab under "My Articles"
		My Articles - Author View	As a profiled Author, I want to view and manage all my articles from a centralised "My Articles" menu, So that I can track the status of each article I have written at any point in time.	**My Articles Menu Availability** ▪ Given that I am logged in as a profiled Author ▪ When I view the CMS navigation ▪ Then a "My Articles" menu item shall be visible and accessible **Published Tab** ▪ Given that I have clicked on "My Articles" ▪ When I select the "Published" tab ▪ Then I shall see a list of all my articles that have been approved by the MAG Admin and are currently live on the Q-Blog **Drafts Tab** ▪ Given that I have clicked on "My Articles" ▪ When I select the "Drafts" tab ▪ Then I shall see a list of all my articles that have been saved as drafts and have not yet been submitted for review or published **Pending Approval Tab** ▪ Given that I have clicked on "My Articles" ▪ When I select the "Pending Approval" tab

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then I shall see a list of all my articles that have been submitted for review and are currently awaiting approval by MAG Admin **Rejected Tab** ▪ Given that I have clicked on "My Articles" ▪ When I select the "Rejected" tab ▪ Then I shall see a list of all my articles that were submitted for review and rejected by the MAG Admin, along with the reason provided for rejection
		My Articles - MAG Admin View	As an MAG Admin, I want to view and manage my own articles from a "My Articles" menu, So that I can track the status of articles I have personally written.	**My Articles Menu Availability** ▪ Given that I am logged in as an MAG Admin ▪ When I view the CMS navigation ▪ Then a "My Articles" menu item shall be visible and accessible **Published Tab** ▪ Given that I have clicked on "My Articles" as an MAG Admin ▪ When I select the "Published" tab ▪ Then I shall see a list of all articles I have personally written and published **Drafts Tab** ▪ Given that I have clicked on "My Articles" as an MAG Admin ▪ When I select the "Drafts" tab ▪ Then I shall see a list of all articles I have personally written and saved as drafts that have not yet been published **No Approval Required** ▪ Given that I am an MAG Admin and I have written an article ▪ When I click the "Publish" button ▪ Then the article shall be published immediately without being routed f approval and shall not appear in any approval queue

Commented [QO5]: A maker-checker control should be in place when an MAG admin writes an article

Commented [OA6R5]: The MAG Admin is already the one approving articles for Authors. If Nkiru for instance is writing an article, why is she approving that same article? Aren't we just creating bottlenecks?

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ And the article shall become visible in the “Published” tab
		All Articles - MAG Admin View	As an MAG Admin, I want to view and manage all articles on the platform regardless of who wrote them, So that I can effectively oversee all content on the Q-Blog.	**All Articles Menu Availability** ▪ Given that I am logged in as an MAG Admin ▪ When I view the CMS navigation ▪ Then an "All Articles" menu item shall be visible and accessible in addition to "My Articles" **Published Tab** ▪ Given that I have clicked on "All Articles" ▪ When I select the "Published" tab ▪ Then I shall see a list of all articles from all Authors and the MAG Admin that are currently live and publicly visible on the Q-Blog **Unpublished Tab** ▪ Given that I have clicked on "All Articles" ▪ When I select the "Unpublished" tab ▪ Then I shall see a list of all articles that were previously published but have since been unpublished by the MAG Admin **Pending Approval Tab** ▪ Given that I have clicked on "All Articles" ▪ When I select the "Pending Approval" tab ▪ Then I shall see a list of all articles submitted by profiled Authors that are currently awaiting my review and approval **Rejected Tab** ▪ Given that I have clicked on "All Articles" ▪ When I select the "Rejected" tab

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then I shall see a list of all articles that have been reviewed and rejected, along with the rejection reasons provided at the time of rejection
		Approval Workflow	As an MAG Admin, I want to review and approve or reject articles submitted by profiled Authors, So that only quality content is published on the Q-Blog.	**Approval Notification via CMS** ▪ Given that a profiled Author has submitted an article for approval ▪ When the article is routed to me as the MAG Admin ▪ Then I shall receive a notification in my CMS notification tab informing me that a new article is pending my review and approval **Approval Notification via Email** ▪ Given that a profiled Author has submitted an article for approval ▪ When the article is routed to me as the MAG Admin ▪ Then I shall also receive an email notification to my registered email address informing me of the pending article awaiting my review **Article Approval** ▪ Given that I am logged in as an MAG Admin and I have opened a pending article for review ▪ When I click the "Approve" button ▪ Then the article shall be immediately published and made publicly visible on the Q-Blog ▪ And the article shall move to the "Published" tab under "All Articles" ▪ Then both the Author and I shall receive a notification via the CMS notification tab and email confirming that the article has been approved and published **Article Rejection** ▪ Given that I am logged in as an MAG Admin and I have opened a pending article for review ▪ When I click the "Reject" button

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then the system shall prompt me to enter a reason for rejection before the rejection is finalised ▪ And when I confirm the rejection ▪ Then the article shall be moved to the "Rejected" tab under "All Articles" ▪ Then both the Author and I shall receive a notification via the CMS notification tab and email confirming that the article has been rejected along with the reason provided **Author Rejection Notification** ▪ Given that my article has been rejected by the MAG Admin as an Author ▪ When the rejection is confirmed ▪ Then I shall receive a notification in my CMS notification tab and an email to my registered email address informing me that my article has been rejected and display the reason provided by the MAG Admin ▪ And the article shall appear in my "Rejected" tab under "My Articles"
7.	**Analytics and Reporting (MAG Admin)**	Google Analytics 4 ("**GA4**") Integration	As an MAG Admin, I want the Q-Blog to be integrated with Google Analytics, So that I can monitor the blog's traffic and performance data on the analytics platform.	**GA4 Integration** ▪ Given that the Q-Blog platform has been deployed ▪ When I log in to the FMDQ Google Analytics 4 account ▪ Then Q-Blog traffic and engagement data shall be visible and trackable within the GA4 dashboard alongside data from other FMDQ-owned websites **Data Accuracy** ▪ Given that the GA4 integration is active ▪ When a public user visits and interacts with the Q-Blog ▪ Then the user's activity including page views, session duration, and traffic source shall be accurately captured and reflected in the GA4 dashboard in real time or within the standard GA4 reporting delay ▪ And this shall be visible only to the MAG Admin

SSD for the Creation of Q-Blog on FMDQ Website

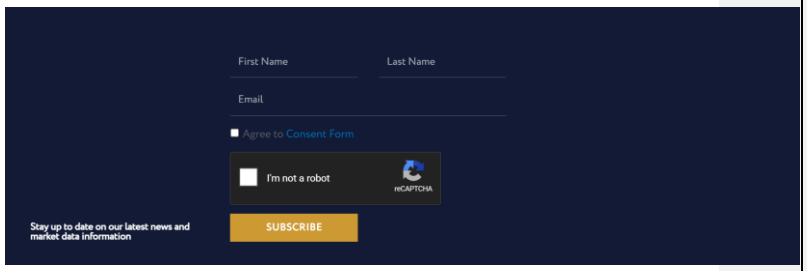
S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
		Native Analytics Dashboard	As an MAG Admin, I want the CMS to have its own native analytics dashboard, So that I can view and monitor Q-Blog performance data directly on the platform and also compare its report with GA4.	**Dashboard Availability** ▪ Given that I am logged in to the CMS as an MAG Admin ▪ When I navigate to the analytics section ▪ Then a native analytics dashboard shall be displayed showing an overview of the Q-Blog's performance metrics **Dashboard Access Restriction** ▪ Given that the native analytics dashboard is available on the CMS ▪ When a profiled Author logs in to the CMS ▪ Then the analytics dashboard shall not be visible or accessible to the Author and shall be restricted to the MAG Admin only
		Traffic Monitoring	As an MAG Admin, I want to track page views and unique visitors for each article, So that I can understand how much readership each piece of content is generating.	**Page Views Tracking** ▪ Given that I am on the native analytics dashboard ▪ When I view the traffic metrics ▪ Then the system shall display the total number of page views recorded for each published article individually **Unique Visitors Tracking** ▪ Given that I am on the native analytics dashboard ▪ When I view the traffic metrics ▪ Then the system shall display the number of unique visitors recorded for each published article individually, distinct from the total page view count **Direct Traffic Sources** ▪ Given that I am on the native analytics dashboard ▪ When I view the traffic source metrics

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then the system shall display the volume of traffic arriving at the Q-Blog via direct visits, where users have navigated to the platform by typing the URL directly into their browser or search engine **Traffic Sources from Social Media** ▪ Given that I am on the native analytics dashboard ▪ When I view the traffic source metrics ▪ Then the system shall display the volume of traffic arriving at the Q-Blog from social media platforms **Traffic Sources through Referral** ▪ Given that I am on the native analytics dashboard ▪ When I view the traffic source metrics ▪ Then the system shall display the volume of referral traffic arriving at the Q-Blog from other sources and links other than search engines and social media platforms
		Engagement Metrics	As an MAG Admin, I want to track reader engagement metrics on the Q-Blog, So that I can understand how readers are interacting with the content beyond just visiting the page.	**Average Reading Time Tracking** ▪ Given that I am on the native analytics dashboard ▪ When I view the engagement metrics ▪ Then the system shall display the average reading time recorded for each published article, reflecting how long readers are spending on each article page on average **Social Share Count Tracking** ▪ Given that I am on the native analytics dashboard ▪ When I view the engagement metrics ▪ Then the system shall display the total number of times each article has been shared across all available social media platforms, by tracking the use of the Share button feature

SSD for the Creation of Q-Blog on FMDQ Website

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
		Content Performance	As an MAG Admin, I want to view content performance dashboards, So that I can identify which articles, categories, and authors are generating the most engagement and readership on the Q-Blog.	**Top Performing Articles** ▪ Given that I am on the native analytics dashboard ▪ When I view the content performance section ▪ Then the system shall display a ranked list of the top-performing articles on the Q-Blog based on page views and engagement metrics **Top Performing Categories** ▪ Given that I am on the native analytics dashboard ▪ When I view the content performance section ▪ Then the system shall display a breakdown of performance by content category, showing which categories are generating the most traffic and engagement **Top Performing Authors** ▪ Given that I am on the native analytics dashboard ▪ When I view the content performance section ▪ Then the system shall display a breakdown of performance by author, showing which authors are generating the most readership and engagement across their published articles **Date Range Filter** ▪ Given that I am on the native analytics dashboard ▪ When I apply a date range filter to the content performance dashboard ▪ Then all metrics displayed shall update to reflect only the data recorded within the selected date range
		Data Export	As an MAG Admin, I want to be able to export analytics data from the native dashboard,	**Export as CSV** ▪ Given that I am on the native analytics dashboard and I am viewing any analytics report ▪ When I select the option to export as CSV

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
			So that I can use the data for reporting, presentations, and stakeholder updates outside of the CMS platform.	▪ Then the system shall generate and download a CSV file containing the report data presented in a structured table format that accurately reflects what is displayed on the dashboard **Export as PDF** ▪ Given that I am on the native analytics dashboard and I am viewing any analytics report ▪ When I select the option to export as PDF ▪ Then the system shall generate and download a PDF file containing the report data as displayed on the dashboard, including all charts, tables, and metrics **Export Access Restriction** ▪ Given that the data export functionality is available on the CMS ▪ When a profiled Author logs in to the CMS ▪ Then the export functionality shall not be visible or accessible to the Author and shall be restricted to the MAG Admin only
8.	**FMDQ Newsletter Subscription**	Subscription Form Display	As a public user, I want to be able to see a newsletter form on all pages of the Q-Blog, So that I can subscribe to the FMDQ newsletter and be updated on the latest news, without having to navigate away from the blog.	**Form Placement** ▪ Given that I am on any page of the Q-Blog ▪ When the page loads ▪ Then the FMDQ newsletter subscription form shall be displayed just above the footer of every Q-Blog page, consistent with its placement on the existing FMDQ website **Form Consistency** ▪ Given that the subscription form is displayed on the Q-Blog ▪ When I compare it to the subscription form on the existing FMDQ website

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then the form shall be identical in appearance, content, and functionality to the existing FMDQ newsletter subscription form including all fields, the consent checkbox, the CAPTCHA verification, and the "Subscribe" button **Kindly refer to the screenshot below:** 
		Subscription Form Fields	As a public user, I want to fill in my details on the subscription form, So that I can successfully subscribe to the FMDQ newsletter.	**Form Fields Display** ▪ Given that I can see the newsletter subscription form above the footer of a Q-Blog page ▪ When the page loads ▪ Then the form shall display the following fields: a. First Name b. Last Name c. Email Address d. "Agree to Consent Form" checkbox e. "I'm not a robot" CAPTCHA verification f. "Subscribe" button. **First Name & Last Name** ▪ Given that I am filling in the subscription form ▪ When I enter my first name and last name in their respective fields

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then the system shall accept and retain the values entered without any errors **Email Address** ▪ Given that I am filling in the subscription form ▪ When I enter a valid email address in the email field ▪ Then the system shall accept the value entered ▪ When I enter an invalid or incorrectly formatted email address ▪ Then the system shall display a clear validation error prompting me to enter a valid email address **Empty Field Submission** ▪ Given that I am filling in the subscription form ▪ When I attempt to submit the form with one or more required fields left empty ▪ Then the system shall display a clear validation error identifying the empty fields and the form shall not be submitted until all required fields are completed
		Consent and CAPTCHA Verification	As a public user, I want to confirm my consent and verify that I am not a robot before subscribing, So that my subscription is compliant with FMDQ's data privacy standards.	**Consent Checkbox** ▪ Given that I am filling in the subscription form ▪ When I attempt to submit the form without checking the "Agree to Consent Form" checkbox ▪ Then the system shall display a validation error informing me that I must agree to the consent form before subscribing and the form shall not be submitted **Consent Form Link** ▪ Given that I can see the "Agree to Consent Form" checkbox on the subscription form ▪ When I click on the "Consent Form" hyperlink

S/N	Module/Screen	Sub-Module	User Story	Acceptance Criteria
				▪ Then I shall be directed to or shown FMDQ's consent form so that I can read its contents before agreeing **CAPTCHA Verification** ▪ Given that I have completed all fields and checked the consent checkbox ▪ When I attempt to submit the form without completing the CAPTCHA verification ▪ Then the system shall display a validation error ▪ And the form shall not be submitted until the CAPTCHA has been successfully completed **CAPTCHA Completion** ▪ Given that I have completed the CAPTCHA verification ▪ When the CAPTCHA is successfully validated ▪ Then the system shall allow me to proceed with submitting the form
		Subscription Action	As a public user, I want to submit the completed subscription form, So that I can begin receiving FMDQ newsletters directly to my inbox.	**Successful Subscription** ▪ Given that I have completed all required fields, checked the consent checkbo and passed the CAPTCHA verification ▪ When I click the "Subscribe" button ▪ Then my details shall be successfully captured and passed to FMDQ's existi newsletter system and a confirmation message shall be displayed acknowledgi my subscription **Duplicate Subscription** ▪ Given that I have previously subscribed to the FMDQ newsletter using a particular email address ▪ When I attempt to submit the form again using the same email address ▪ Then the system shall display an appropriate message informing me that the email address is already registered as a subscriber to the FMDQ newsletter

Commented [KA7]: Does the newsletter have an unsubscription link? if yes does it take out the users email/delete the email from the subscribers mailing list.

Commented [OA8R7]: Kindly refer to the comment below.

Commented [QO9]: What is the unsubscription protocol?

Commented [OA10R9]: This is not a functional requirement.

6 Document Information/Approvals for iQx Consult Limited

Document Name	Solution Specification Document
Approved By *Emmanuel Alao*	---------- ***Supervising Head, iQx Consult*** **May 2026**

7 TECHNICAL REQUIREMENTS

The following technical requirements define the infrastructure, architecture, and performance standards necessary to ensure the Q-Blog operates securely, reliably, and at scale:

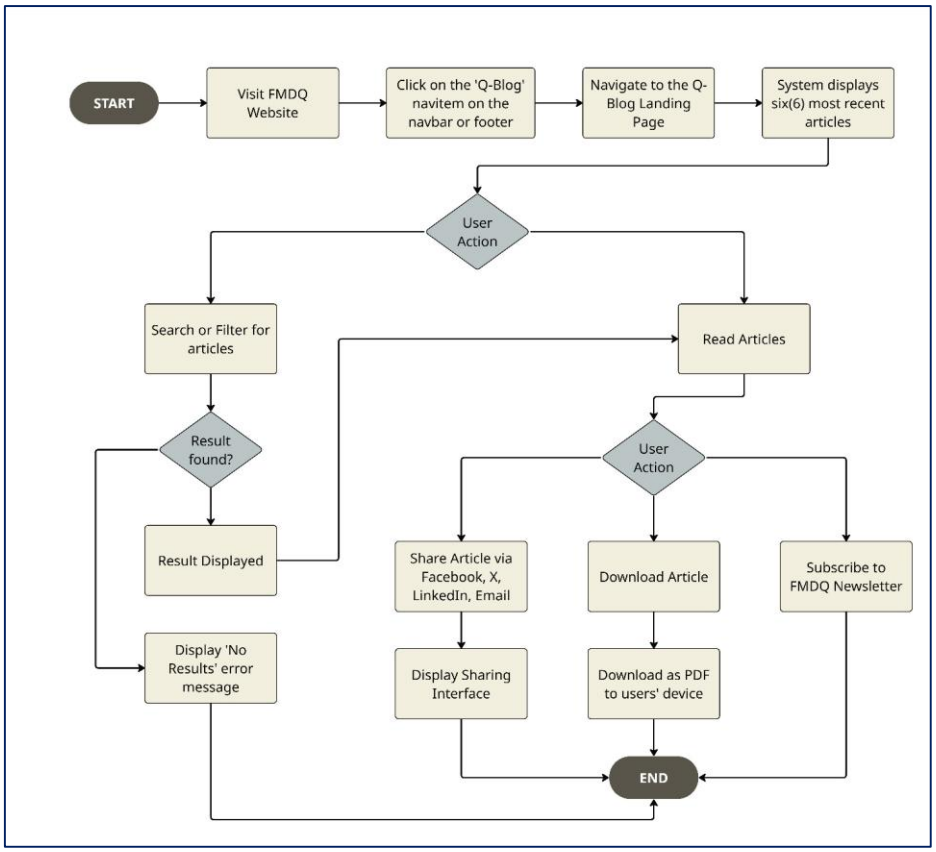
- **Database Management**: The system shall use a reliable and scalable Database Management System ("**DBMS**") to store and manage visitor records, user accounts, booking data, audit logs, and configuration parameters. The DBMS shall support data backup, replication, and performance optimisation strategies
- **Security Measures**: The system shall implement robust security controls including data encryption, secure access controls, user authentication, role-based permissions, firewall configurations, and regular security audits to protect visitors' data and system integrity
- **Performance**: The system shall be designed to handle concurrent usage by multiple user types across multiple touchpoints without significant latency or system degradation
- **Scalability**: The system shall be built to accommodate growth in user accounts, visitor volumes, and data storage without requiring significant infrastructure redesign
- **Availability**: The system shall maintain high availability during FMDQ's operational hours, with planned maintenance scheduled outside business hours
- **Usability**: The system shall feature a user-friendly interface optimised for ease of use across Security gate devices (tablets) and reception desks
- **Maintainability**: The system shall be developed with modular architecture to facilitate easy bug fixes, configuration changes, and future feature enhancements
- **Compatibility**: The system shall be compatible with major popular browsers (Google Chrome, Microsoft Edge, Mozilla Firefox) and responsive across desktop and tablet form factors
- **Testing**: The system shall undergo comprehensive testing including unit testing, integration testing, user acceptance testing (UAT), and security/penetration testing prior to go-live
- **Notification Integration**: The system shall integrate with FMDQ's Microsoft 365 environment to support automated email and Microsoft Teams notifications across the visitor lifecycle
- **Error Handling**: The system shall implement robust error handling mechanisms, providing meaningful user-facing messages without exposing internal system state or stack trace information

SSD for the Creation of Q-Blog on FMDQ Website

7.1 High-Level Visual Model

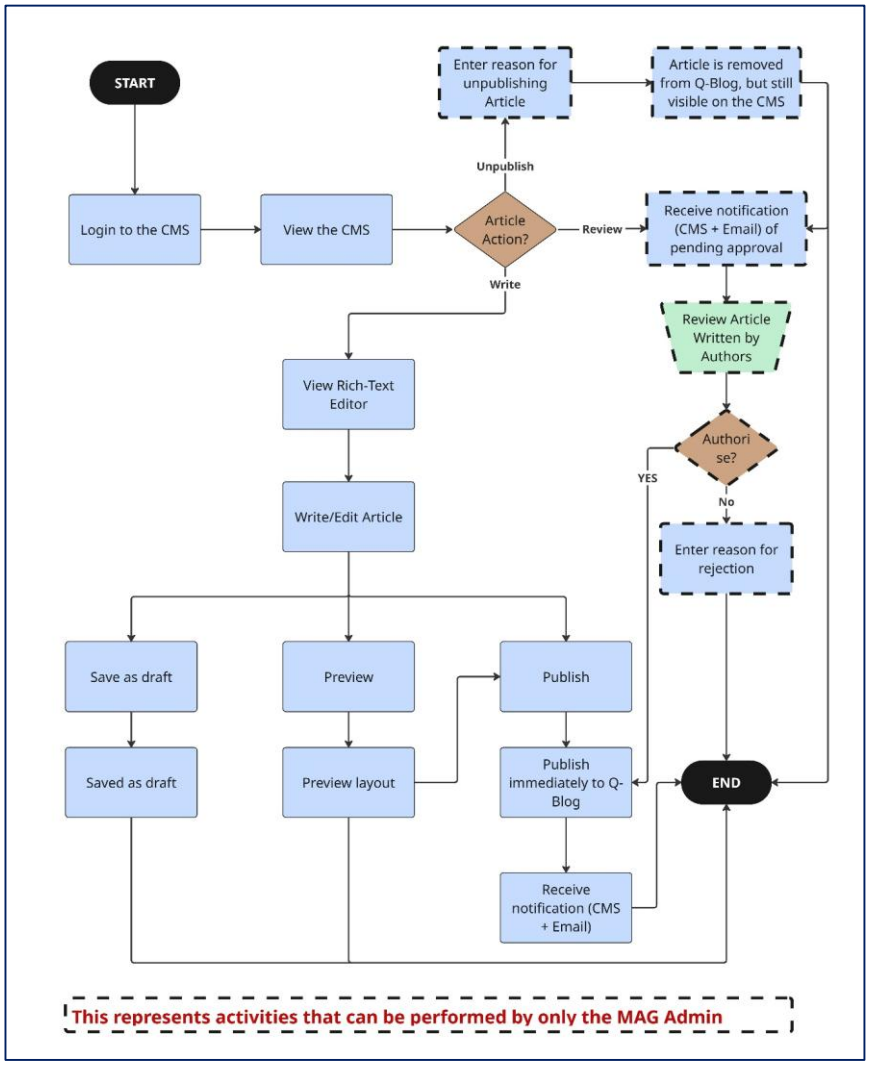
a. Public User Journey to Read Articles

The Q-Blog is accessible to all visitors of the FMDQ Group website via a dedicated hyperlink titled "FMDQ Q-Blog" placed in the main navigation menu beside the "FMDQ Next" item and in the footer across all website pages. Users can search for articles by keyword, filter by category or author, and sort results to find content relevant to their interests.



b. CMS User Journey (Articles)

The Q-Blog CMS is accessible exclusively to both the MAG Admin and profiled Authors. Both users log in with secure credentials and are presented with role-specific interfaces. A profiled Author can create articles using a rich-text editor, preview the finished layout, save drafts for later, or submit an article for approval. Once submitted, the article is routed to the MAG Admin for approval. The Author is notified via CMS and email once a decision is made, and can view the outcome, along with any rejection reason. The MAG Admin, in addition to creating and publishing articles independently without approval, oversees all content on the platform. Upon receiving a pending article notification, the MAG Admin can review, approve, or reject the submission, with both parties automatically notified of the outcome.



7.2 Logical Data Model/Data Dictionary

The below data requirements describe the business data needed by the application system.

Entities:

- **User:** This entity stores information about all registered users on the CMS platform regardless of their role
- **Author Profile:** This entity stores publicly visible profile information for users designated as Authors on the platform
- **Article:** This entity stores all information related to blog articles created and published on Q-Blog
- **Category:** This entity stores the predefined content categories under which articles are organised
- **Article_Tag:** This is a junction entity that manages the many-to-many relationship between articles and tags
- **Media:** This entity stores all media assets uploaded to the CMS for use within articles
- **Approval Workflow:** This entity tracks the review and approval history of each article
- **Subscriber:** This entity stores information about users who subscribe to the FMDQ newsletter via the Q-Blog platform
- **Analytics:** This entity stores on-platform engagement data for each published article

Relationships:

S/N	Relationship	Type	Description
1	User to Author Profile	One-to-One	A user may have one author profile
2	Author Profile to Article	One-to-Many	An author can write multiple articles
3	Category to Article	One-to-Many	A category can contain multiple articles
4	Article to Article Tag	One-to-Many	An article can have multiple tags
5	Tag to Article Tag	One-to-Many	A tag can be associated with multiple articles
6	Article to Media	One-to-Many	An article can contain multiple media assets
7	Article to Approval Workflow	One-to-One	An article can have one approval workflow
8	Article to Analytics	One-to-Many	An article can have multiple analytics records
9	User to Approval Workflow	One-to-Many	A user can action multiple workflow records

7.3 Interface Requirements

Hardware Interfaces

The system should be able to perform with the following hardware interface requirements:

- Processor speed of 1.0ghz or more for tablet devices
- Processor speed of 1.5ghz or more for desktop and computer gadgets

- Ram of 500mb and above for all devices
- Free storage memory capacity of more than 500mb on each device

Software Interfaces

The system should have a smooth performance on the following operating systems and internet browsers:

- Windows/ Android/ Linux/ Mac / Chrome or any other operating system
- Mozilla Firefox / Google chrome / opera mini / Edge / UC browser or internet explorer and any other browsers

Communications Interfaces

- Internet connectivity will be required for communication to occur on the system

7.4 Data Conversion Requirements

- **Media File Conversion**: All images and media assets uploaded to the CMS must be automatically converted to web-optimised formats (e.g., JPEG, PNG, WebP) to ensure fast load times and consistent rendering across all devices and browsers
- **Document to PDF Conversion**: The system must support the conversion of published articles to downloadable PDF format upon request by a public user, preserving the article's formatting, images, and metadata in the output file
- **Analytics Data Export Conversion**: Analytics data generated by the internal CMS dashboard must be convertible and exportable into standard formats including CSV and PDF for reporting and presentation purposes
- **Rich Text to HTML Conversion**: Content created in the CMS rich-text editor must be accurately converted to HTML for rendering on the public-facing blog pages, with all formatting including headings, bullet points, hyperlinks, and embedded media preserved
- **Newsletter Subscriber Data**: Subscriber email addresses collected via the Q-Blog subscription form must be converted and passed to FMDQ's existing newsletter system in a compatible format to ensure seamless integration without data loss
- **URL Slug Generation**: Article titles entered in the CMS must be automatically converted into SEO-friendly URL slugs (e.g., "Understanding Nigeria's Bond Market" ➔ /blog/understanding-nigerias-bond-market), with special characters and spaces handled appropriately
- **Data Security**: Sensitive data processed during migration shall be handled with appropriate access controls and encryption

7.5 Hardware/Software Requirements

All specifications applied in the interface requirements will be applied here also

7.6 Operational Requirements

The following requirements should be considered for system efficiency:

- **System Availability and Reliability**: The blog should be always available and reliable, with a high uptime percentage. Downtime for maintenance and updates should be scheduled during off-peak hours and communicated in advance
- **User Management**: The system should provide functionality for user management, including the ability for the MAG Super Admin to create, modify, deactivate, and reactivate user accounts and role assignments on the platform as required
- **Data Security and Privacy**: The system should ensure data security and privacy and other sensitive information. It should implement appropriate security measures, access controls, data encryption, and comply with relevant data protection regulations
- **Scalability and Performance**: The system should be designed to handle a growing number of users. It should be scalable and perform efficiently even with increased system usage
- **Backup and Recovery**: Regularly back up the system's data and configurations. Establish a comprehensive disaster recovery plan to swiftly restore the system in case of data loss or system failure
- **Monitoring and logging**: The system should be designed to provide real-time monitoring and logging of system activities to enable system administrators to identify and address issues promptly
- **Compliance and Regulatory Requirements**: All user actions within the CMS including content creation, edits, approvals, and publications must be logged and traceable for audit purposes
- **Backup and recovery**: The system should be designed to perform regular backups and implement recovery mechanisms in line with the FMDQ Data Recovery policy, to ensure that data is not lost in case of system failures

Commented [KA11]: Should we not specify the frequency? Please clarify.

Commented [OA12R11]: This refers to the fmdq data backup and recovery policy.

7.6.1 Security and Privacy

The following security and privacy requirements are critical to ensure the confidentiality, integrity, and availability of the Q-Blog. The system must comply with applicable data privacy regulations including the Nigeria Data Protection Act ("**NDPA**"), Nigeria Data Protection Regulation ("**NDPR**"), and relevant FMDQ internal data protection policies. The system must also adhere to industry best practices such as the OWASP Application Security Verification Standard version 4

Data Encryption:

- Ensure that all Personally Identifiable Information ("**PII**") data, both in transit and at rest use strong and secure encryption algorithms, such as Advanced Encryption Standard ("**AES**") or Twofish, to encrypt sensitive data. Avoid using weak encryption algorithms that can be easily cracked
- **Encrypt Data in Transit**: Use Transport Layer Security ("**TLS**") to encrypt data in transit between the web application and the client browser. This helps protect against eavesdropping and man-in-the-middle attacks
- **Encrypt Data at Rest**: Store sensitive data in an encrypted format in the database or on disk. This helps protect against data breaches and unauthorised access to the data

- **Use Key Management Best Practices**: Implement key management best practices, such as key rotation, to ensure that encryption keys are kept secure and are not compromised. Avoid hardcoding encryption keys in the application code or configuration files
- **Access Control:** Implement strict access controls to limit data access to authorised personnel only. Role-based access control ("**RBAC**") should be employed, granting different levels of access based on the principle of least privilege
- **Consent and Transparency:** Visitors shall be informed of the purpose of data collection at the point of pre-registration or self-service portal submission, in compliance with NDPA requirements
- **Data Minimisation:** The system shall collect and retain only the PII necessary for visitor management operations. Unnecessary sensitive data shall not be stored
- **Data Retention and Deletion:** Define clear policies for data retention and deletion. Regularly review and delete personal data that is no longer required for HR purposes, adhering to applicable laws and regulations
- **Employee Training and Awareness:** Provide regular training to staff and security personnel on data privacy best practices, security protocols, and their responsibilities in handling PII. Promote a culture of data privacy awareness throughout the organisation

    - **Input and Output Handling:**
        - All database queries shall use parameterised SQL statements to prevent SQL injection attacks
    - **Session Management:**
        - **Session Timeout**: Implement session timeout to ensure that inactive sessions are terminated after a specified time. This helps prevent session hijacking attacks by closing the session after a certain period of inactivity.
        - **Secure Session ID Generation**: Generate random and unpredictable session IDs to prevent session fixation attacks. Session IDs should be unique for each user and should not be guessable or predictable.
        - **Session Encryption**: Encrypt session data and the session ID to prevent eavesdropping and session hijacking attacks. Use SSL/TLS to encrypt session data in transit and store session data in an encrypted format in the server.
        - **Cookie Security**: Use secure cookies to transmit session data between the client and server. Secure cookies should have the 'secure' flag set to prevent interception by attackers, and the 'httpOnly' flag set to prevent client-side scripting attacks.
        - **Session Revocation**: Implement session revocation mechanisms to terminate sessions that are suspected to be compromised or hijacked. This helps prevent further exploitation of the session by attackers.
    - **Software versions and updates**: At a minimum, the software used in and around the application must be up-to-date, and there must be no known vulnerabilities
    - **Secure Coding:** The application must be developed using secure coding practices to prevent common security vulnerabilities, such as SQL injection, cross-site scripting (XSS), and buffer overflow attacks
    - **Vulnerability Assessment and Penetration Testing (VAPT):** VAPT must be done against the OWASP Top 10 Web Application Vulnerabilities (Version 2021), and all identified vulnerabilities must be remediated

7.6.2 Audit Trail

The system shall maintain a comprehensive and tamper-proof log of all user activities performed within the CMS, including content creation, edits, submissions, approvals, publications, and account management actions. All audit logs must be accessible exclusively to iQx and retained for a minimum period as defined by FMDQ Group's internal data governance and compliance policies.

7.6.3 Reliability

The system should be able to function correctly and consistently, without any unexpected failures or errors. Regular system maintenance and testing should be conducted to identify and address any potential issues before Go-Live.

7.6.4 Data Backup and Recoverability

The application must have a robust backup and recovery system in place to ensure that data can be restored in the event of a system failure or data breach

- If the system is unavailable to users (experiencing downtime) because of a system failure, the failure must be detected, and function will be restored within thirty (30) minutes to one (1) hour
- In the event the database is corrupted, the database must be capable of being restored to its condition of no more than one (1) hour before the corruption occurred

7.6.5 General Performance

- **Response time for queries and updates**: The system should be able to respond to user queries and updates within a few seconds to ensure a seamless user experience. However, certain queries or updates may require more processing time, such as bulk updates. In such cases, the system should provide feedback to the user, indicating that the query or update is being processed and the estimated time for completion
- **Expected rate of user activity**: The system will experience consistent need-based user activity. User activity will be monitored and analysed by the System Administrators

7.6.6 Capacity

The system will be mainly utilised by the MAG and external users who need to view financial market articles. The system should be built to handle a large volume of users.

7.6.7 Data Retention

The data retention policy shall conform to applicable regulatory and internal FMDQ requirements. PII and related records shall be retained for a period of seven (**7**) years from the date of creation. Upon expiry of the retention period, data shall be securely archived or permanently deleted/anonymised in a manner compliant with NDPR/A provisions. The system shall provide configurable retention period settings, manageable by the Super Administrator. All data archival and destruction activities shall be logged.

7.6.8 Error Handling and Logging

Error messages presented to users shall be generic and informative without revealing internal application state, file paths, stack traces, or database structure. Custom error messages shall replace default framework-generated messages. All errors shall be logged internally with full technical detail for System Administrator review. Error logs shall be reviewed periodically to identify recurring issues requiring remediation.

7.6.9 Validation Rules

All mandatory fields identified in the Functional Requirements (Section 4.4) shall be enforced as required at the point of data entry, with clear validation messages displayed to users upon incomplete submission. Business rules, including approval workflow routing, badge inventory limits, and escalation thresholds shall be enforced at the application layer and shall be configurable by the System Administrator where applicable.

Commented [KA13]: Please confirm that all downtime shall be reported in line with the Firm's incident reporting.

Commented [OA14R13]: iQx is expected to report all system's downtime in their incident reporting. BUG cannot verify or validate the incident reports as we have no oversight of it.

Commented [KA15]: Could you please specify which Unit/Group/Division that is responsible for this.

Commented [OA16R15]: The implementation team(build or buy) is not determined by BUG.

7.6.10 Conventions/Standards

- The system shall comply with the Nigeria Data Protection Act (NDPA) and Nigeria Data Protection Regulation (NDPR), and the General Data Protection Regulation ("**GDPR**")
- The system shall align with FMDQ's internal IT standards, branding guidelines, and accessibility requirements
- The system should adhere to industry-standard security protocols and frameworks